

Lecture Set 4 Supplement

DNSC 8328

Reading Conditional Independencies from a DAG:
The Moral Graph Criterion

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THE MORAL GRAPH CRITERION

- ▶ Given a DAG $\mathcal{G}$ on the variables $Y = \{Y_1, \dots, Y_I\}$, we want to decide whether

$$X \perp W \mid Z$$

holds for disjoint subsets X, W, Z of Y in *every* distribution that factorizes as $p(\mathbf{y}) = \prod_{i=1}^I p(y_i \mid \Pi_i)$.

- ▶ Lauritzen, Dawid, Larsen and Leimer (1990) showed that this can be answered by an ordinary *separation* question in an *undirected* graph built from $\mathcal{G}$.
- ▶ **Theorem.** $X \perp W \mid Z$ is implied by the DAG if and only if Z separates X from W in the moral graph of the ancestral set of $X \cup W \cup Z$.
- ▶ The criterion is equivalent to *d-separation* of Pearl (1988); both give exactly the same answers.

DEFINITIONS

- ▶ **Ancestral set** $An(S)$: the nodes in S together with all their ancestors (parents, parents' parents, ...). The *ancestral graph* $\mathcal{G}_{An(S)}$ is the sub-DAG on these nodes.
- ▶ **Moral graph** $\mathcal{G}^m$: obtained from a DAG by
 1. joining (“marrying”) every pair of nodes that have a common child, and
 2. dropping the directions of all arcs.
- ▶ **Separation**: in an undirected graph, Z *separates* X from W if every path from a node in X to a node in W passes through a node in Z .

Equivalently, X and W are disconnected once the nodes in Z and their edges are removed.

THE ALGORITHM

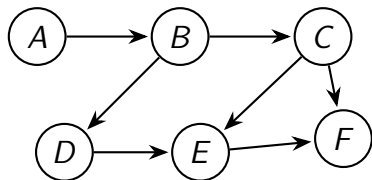
To check whether $X \perp W \mid Z$:

1. **Ancestral graph.** Keep only the nodes in $X \cup W \cup Z$ and their ancestors; delete all other nodes and their arcs.
2. **Moralize.** For every node of this graph, join each pair of its parents by an undirected edge (if not already adjacent).
3. **Drop directions.** Replace every arc by an undirected edge.
4. **Delete Z .** Remove the conditioning nodes and all their edges.

Conclusion:

- ▶ If no path connects X and W , then $X \perp W \mid Z$.
- ▶ If a path remains, the DAG does *not* imply the independence.

RUNNING EXAMPLE



$$\begin{aligned}\Pi_A &= \emptyset, \quad \Pi_B = \{A\} \\ \Pi_C &= \{B\}, \quad \Pi_D = \{B\} \\ \Pi_E &= \{C, D\}, \quad \Pi_F = \{C, E\}\end{aligned}$$

$$p(A, \dots, F) = p(A) p(B|A) p(C|B) p(D|B) p(E|C, D) p(F|C, E)$$

► Colors used in the following slides:



first set



second set



conditioning set



moral edge

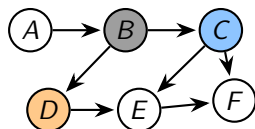


deleted in Step 4

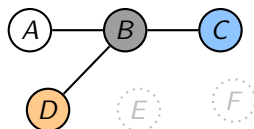


not in ancestral set

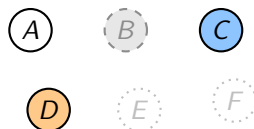
EXAMPLE 1: IS $C \perp D \mid B$?



Step 0: the DAG



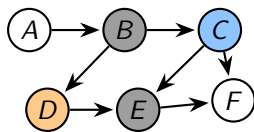
Steps 1-3: moral ancestral graph



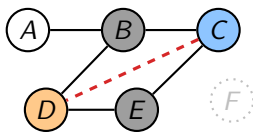
Step 4: delete $\{B\}$

- ▶ Ancestral set of $\{B, C, D\}$ is $\{A, B, C, D\}$: E and F are dropped.
- ▶ No two nodes in the ancestral graph share a child $\Rightarrow$ no moral edges.
- ▶ After deleting B , C and D are disconnected $\Rightarrow C \perp D \mid B$.

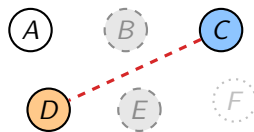
EXAMPLE 2: IS $C \perp D \mid \{B, E\}$?



Step 0: the DAG



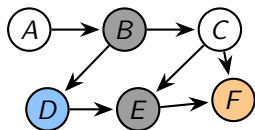
Steps 1–3: moral ancestral graph



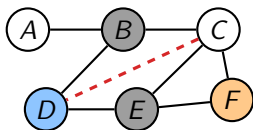
Step 4: delete $\{B, E\}$

- ▶ Now $E \in An(\{B, C, D, E\})$, so its parents C and D are married.
- ▶ After deleting B and E , the moral edge $C - D$ remains.
- ▶ C and D are **not** separated $\Rightarrow C \not\perp D \mid \{B, E\}$ (“explaining away”).

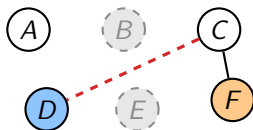
EXAMPLE 3: IS $D \perp F \mid \{B, E\}$?



Step 0: the DAG



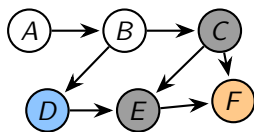
Steps 1-3: moral ancestral graph



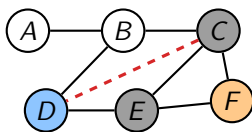
Step 4: delete $\{B, E\}$

- ▶ The ancestral set is the whole graph; the moral edge is $C - D$ (C and E are already adjacent).
- ▶ After deleting B and E , the path $D - C - F$ remains.
- ▶ $\Rightarrow D \not\perp F \mid \{B, E\}$: conditioning on E opened a path through C .

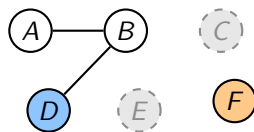
EXAMPLE 4: IS $D \perp F \mid \{C, E\}$?



Step 0: the DAG



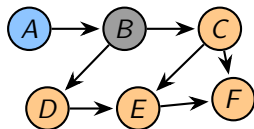
Steps 1-3: moral ancestral graph



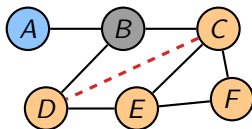
Step 4: delete $\{C, E\}$

- ▶ Same moral graph as Example 3, but now C and E are deleted.
- ▶ F has no remaining neighbours $\Rightarrow D \perp F \mid \{C, E\}$.
- ▶ In fact $F \perp \{A, B, D\} \mid \{C, E\}$: the local Markov property at F , whose parents are $\{C, E\}$.

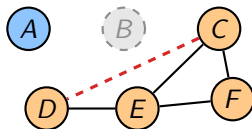
EXAMPLE 5: IS $A \perp \{C, D, E, F\} \mid B$?



Step 0: the DAG



Steps 1-3: moral ancestral
graph



Step 4: delete $\{B\}$

- ▶ The ancestral set is the whole graph, with moral edge $C - D$.
- ▶ Deleting B isolates A , since B is its only neighbour.
- ▶ $\Rightarrow A \perp \{C, D, E, F\} \mid B$; adding D to the conditioning set changes nothing.

SUMMARY OF THE EXAMPLES

Query	Ancestral set	Moral edges	Result
C vs $D \mid B$	$\{A, B, C, D\}$	none	$C \perp D \mid B$
C vs $D \mid \{B, E\}$	$\{A, B, C, D, E\}$	$C - D$	dependent
D vs $F \mid \{B, E\}$	all	$C - D$	dependent
D vs $F \mid \{C, E\}$	all	$C - D$	$D \perp F \mid \{C, E\}$
A vs $\{C, D, E, F\} \mid B$	all	$C - D$	$A \perp \{C, D, E, F\} \mid B$

- ▶ All results agree with d-separation. In R, `bnlearn` gives the same answers for all 480 possible queries on this DAG (every pair of nodes, every conditioning set).
- ▶ Conditioning can *remove* dependence (Example 4) or *create* it (Examples 2 and 3).

CHECKING IN R WITH bnlearn

```
library(bnlearn)
dag <- model2network("[A] [B|A] [C|B] [D|B] [E|C:D] [F|C:E] ")

# Steps 1-3 for the query D vs F given {B, E}
S   <- c("D", "F", "B", "E")
anc <- unique(c(S, unlist(lapply(S, ancestors, x = dag))))
m   <- moral(subgraph(dag, anc))      # moral, undirected
arcs(m)                               # includes C - D

# d-separation, for comparison
dsep(dag, "D", "F", c("B", "E"))     # FALSE: dependent
dsep(dag, "D", "F", c("C", "E"))     # TRUE:  independent
```